Vivante Programming:

Neural Network

OVXLIB Operation Support

with NPU V9.2.0

Document Revision 0.52

18 December 2023

VERISILICON

Legal Notices

Table of Contents

1 Overview

This document summarizes the neural network OVXLIB operations supported by Vivante NPU V9.2.0.

For the operations supported by NPU V7.x, V8.0.x, V8.1.x, and V9.0.x, see *Vivante Programming: Neural Network OVXLIB Operation Support with NPU*.

For the operations supported by NPU V9.1.x, see *Vivante Programming: Neural Network OVXLIB Operation Support with NPU V9.1.x*.

1.1 Hardware Requirements

Compatible NPU hardware must have the parallel processing unit (PPU) equipped.

1.2 Software Requirements

Neural network OVXLIB operations require the following compatible software stacks to perform inference on the NPU hardware:

- Vivante GALVIP unified driver v6.4.17 or later
- OVXLIB v1.2.2 or later
  This high-level wrapper library has additional NN APIs for OpenVX.

2 OVXLIB Operation Support

This chapter lists OVXLIB operations supported by NPU V9.2.0. Operations are categorized by function in the following sections:

- *2.2, Basic Operations*
- *2.3, Activation Operations*
- *2.4, Elementwise Operations*
- *2.5, Normalization Operations*
- *2.6, Reshape Operations*
- *2.7, RNN Operations*
- *2.8, Pooling Operations*
- *2.9, Miscellaneous Operations*

2.1 Abbreviations in This Chapter

Abbreviations used in the following tables are listed as follows:

- Data type abbreviations:
  - **asym-u8** asymmetric_affine-uint8
  - **asym-i8** asymmetric_affine-int8
  - **pc-sym-i8** perchannel_symmetric-int8
  - **fp32** float32
  - **fp16** float16
  - **bool8** bool8
  - **sym-i16** symmetric-int16
  - **int32** int32
  - **bf16** bfloat16
- Execution engine abbreviations:
  - **NN** Vivante **N**eural-**N**etwork Engine
  - **PPU** Vivante **P**arallel **P**rocessing **U**nit

2.2 Basic Operations

Table 1 lists the basic OVXLIB operations supported by the NPU.

Note 1: The support for execution engines may vary with software releases.
Note 2: Some operations may not be supported by certain NPU hardware versions.

Table 1. Basic Operation Support with NPU

Operation	Tensor			Execution Engine	
	Input	Kernel	Output	NN	PPU
VSI_NN_OP_CONV1D	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_CONV2D	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_CONV3D	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_DECONVOLUTION	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	

Operation	Tensor			Execution Engine	
	Input	Kernel	Output	NN	PPU
VSI_NN_OP_DECONVOLUTION1D	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	Supported
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_FCL2	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_GROUPED_CONV1D	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	Supported
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_GROUPED_CONV2D	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	Supported
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	

2.3 Activation Operations

Table 2 lists the OVXLIB activation operations supported by the NPU.

Note 1: The support for execution engines may vary with software releases.
Note 2: Some operations may not be supported by certain NPU hardware versions.
Note 3: In the table, operations in blue are either added or updated in this release.

Table 2. Activation Operation Support with NPU

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_ABS	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_ACOSH	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_ATAN	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_ATANH	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_CELU	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_CLIP	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_COS	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_ELU	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_ERF	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_EXP	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_GELU	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_HARD_SIGMOID	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_INVERSE_SIGMOID	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_LEAKY_RELU	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_LINEAR	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_LOG	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_LOG_SOFTMAX	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_MISH	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_NEG	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_PRELU	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_RCP	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_RELU	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_RELUN	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_RSQRT	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_SELU	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_SIGMOID	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_SIGN	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_SIN	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_SOFTMAX	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_SOFTRELU	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_SOFTSIGN	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_SQRT	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_SQUARE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_SWISH	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_TAN	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_TANH	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

2.4 Elementwise Operations

Table 3 lists the OVXLIB elementwise operations supported by the NPU.

Note 1: The support for execution engines may vary with software releases.
Note 2: Some operations may not be supported by certain NPU hardware versions.

Table 3. Elementwise Operation Support with NPU

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_ADD	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_ADDN	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_DIVIDE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_FLOORDIV	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_LOGICAL_NOT	bool8	bool8	Supported	
VSI_NN_OP_LOGICAL_OPS	bool8	bool8	Supported	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_MATRIXMUL	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_MAXIMUM	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_MINIMUM	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_MOD	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_MULTIPLY	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_POW	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_RELATIONAL_OPS	asym-u8	bool8	Supported	
	asym-i8	bool8	Supported	
	fp32	bool8	Supported	
	fp16	bool8	Supported	
	bool8	bool8	Supported	
	sym-i16	bool8	Supported	
	bf16	bool8	Supported	
VSI_NN_OP_SELECT	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	bool8	bool8	Supported	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_SUBTRACT	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

2.5 Normalization Operations

Table 4 lists the OVXLIB normalization operations supported by the NPU.

Note 1: The support for execution engines may vary with software releases.
Note 2: Some operations may not be supported by certain NPU hardware versions.

Table 4. Normalization Operation Support with NPU

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_BATCH_NORM	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_BATCHNORM_SINGLE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_GROUP_NORM	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_INSTANCE_NORM	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_L2_NORMALIZE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_LAYER_NORM	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_LPNORM	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_LRN2	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_MOMENTS	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

2.6 Reshape Operations

Table 5 lists the OVXLIB reshape operations supported by the NPU.

Note 1: The support for execution engines may vary with software releases.
Note 2: Some operations may not be supported by certain NPU hardware versions.

Table 5. Reshape Operation Support with NPU

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_ARGMAX	asym-u8	asym-u8 / sym-i16 / int32		Supported
	asym-i8	asym-u8 / sym-i16 / int32		Supported
	fp32	int32		Supported
	fp16	asym-u8 / sym-i16 / int32		Supported
	sym-i16	asym-u8 / sym-i16		Supported
	bf16	asym-u8 / sym-i16		Supported
VSI_NN_OP_ARGMIN	asym-u8	asym-u8 / sym-i16 / int32		Supported
	asym-i8	asym-u8 / sym-i16 / int32		Supported
	fp32	int32		Supported
	fp16	asym-u8 / sym-i16 / int32		Supported
	sym-i16	asym-u8 / sym-i16		Supported
	bf16	asym-u8 / sym-i16		Supported
VSI_NN_OP_BATCH2SPACE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_CONCAT	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_DEPTH2SPACE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_EXPAND_BROADCAST	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_PAD2	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_PERMUTE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_REDUCE	asym-u8	asym-u8	Supported	Supported
	asym-i8	asym-i8	Supported	Supported
	fp32	fp32	Optional	Supported
	fp16	fp16	Optional	Supported
	sym-i16	sym-i16	Optional	Supported
	bf16	bf16	Optional	Supported
VSI_NN_OP_REDUCEL2	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_REORG	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_RESHAPE2	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_REVERSE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_SHUFFLECHANNEL	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_SLICE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_SPACE2BATCH	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_SPACE2DEPTH	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Optional	
VSI_NN_OP_SPLIT	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_SQUEEZE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_STACK	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	
VSI_NN_OP_STRIDED_SLICE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_UNSTACK	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Supported	
	fp16	fp16	Supported	
	sym-i16	sym-i16	Supported	
	bf16	bf16	Supported	

2.7 RNN Operations

Table 6 lists the OVXLIB recurrent neural network (RNN) operations supported by the NPU.

Note 1: The support for execution engines may vary with software releases.
Note 2: Some operations may not be supported by certain NPU hardware versions.

Table 6. RNN Operation Support with NPU

Operation	Tensor			Execution Engine	
	Input	Kernel	Output	NN	PPU
VSI_NN_OP_CONV2D_LSTM	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32	Optional	Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_CONV2D_LSTM_CELL	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32	Optional	Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_GRU	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32	Optional	Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_GRUCELL	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32	Optional	Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_LSTM_OVXLIB	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32	Optional	Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	

Operation	Tensor			Execution Engine	
	Input	Kernel	Output	NN	PPU
VSI_NN_OP_LSTMUNIT_OVXLIB	asym-u8	asym-u8	asym-u8	Supported	
	asym-i8	pc-sym-i8	asym-i8	Supported	
	fp32	fp32	fp32	Optional	Supported
	fp16	fp16	fp16	Optional	
	sym-i16	sym-i16	sym-i16	Optional	
	bf16	bf16	bf16	Optional	
VSI_NN_OP_SVDF	asym-u8	asym-u8	asym-u8	Supported	Supported
	asym-i8	pc-sym-i8	asym-i8	Supported	Supported
	fp32	fp32	fp32		Supported
	fp16	fp16	fp16	Optional	Supported
	sym-i16	sym-i16	sym-i16	Optional	Supported
	bf16	bf16	bf16	Optional	Supported

2.8 Pooling Operations

Table 7 lists the OVXLIB pooling operations supported by the NPU.

Note 1: The support for execution engines may vary with software releases.
Note 2: Some operations may not be supported by certain NPU hardware versions.

Table 7. Pooling Operation Support with NPU

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_AVG_POOL3D	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_GLOBALLPPOOL	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_LPPOOL	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_MAX_POOL3D	asym-u8	asym-u8	Supported	Supported
	asym-i8	asym-i8	Supported	Supported
	fp32	fp32		Supported
	fp16	fp16	Optional	Supported
	sym-i16	sym-i16	Optional	Supported
	bf16	bf16	Optional	Supported
VSI_NN_OP_MAXPOOLWITHARGMAX	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_MAXUNPOOL	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_POOL	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_POOLWITHARGMAX	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_ROI_POOL	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_UPSAMPLE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

2.9 Miscellaneous Operations

Table 8 lists other OVXLIB operations supported by the NPU.

Note 1: The support for execution engines may vary with software releases.
Note 2: Some operations may not be supported by certain NPU hardware versions.
Note 3: In the table, operations in blue are either added or updated in this release.

Table 8. Miscellaneous Operation Support with NPU

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_BUCKETIZE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_CAST	asym-u8 asym-i8 bool8 sym-int16 int32 fp16 bf16 fp32	asym-u8 asym-i8 bool8 sym-int16 int32 fp16 bf16 fp32	Supported	Supported
VSI_NN_OP_CEIL	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
VSI_NN_OP_CONCATSHIFT	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
VSI_NN_OP_CROP_AND_RESIZE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_CUMSUM	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_DATACONVERT	asym-u8 / asym-i8 / sym-i16	asym-u8 / asym-i8 / sym-i16	Supported	
	asym-u8 / asym-i8 / sym-i16	fp16 / fp32 / bf16	Optional	
	fp16 / fp32 / bf16	asym-u8 / asym-i8 / sym-i16 / fp16 / fp32 / bf16	Optional	
VSI_NN_OP_DROPOUT	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	
VSI_NN_OP_EMBEDDING_LOOKUP	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_FLOOR	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_GATHER	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_GATHER_ELEMENTS	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_GATHER_ND	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_GRID_SAMPLE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_ONE_HOT	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
VSI_NN_OP_PROPOSAL	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_REPEAT	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
VSI_NN_OP_RESIZE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_RESIZE_1D	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_RESIZE_3D	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_REVERSESEQUENCE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_ROUND	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16		Supported

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_SCATTER_ELEMENTS	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_SCATTER_ND	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_SCATTER_ND_UPDATE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported
VSI_NN_OP_SEQUENCE_MASK	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
VSI_NN_OP_SIGNAL_FRAME	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
VSI_NN_OP_TILE	asym-u8	asym-u8		Supported
	asym-i8	asym-i8		Supported
	fp32	fp32		Supported
	fp16	fp16		Supported
	sym-i16	sym-i16		Supported
	bf16	bf16		Supported

Operation	Tensor		Execution Engine	
	Input	Output	NN	PPU
VSI_NN_OP_UPSAMPLESCALE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32		Supported
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	Supported
VSI_NN_OP_VARIABLE	asym-u8	asym-u8	Supported	
	asym-i8	asym-i8	Supported	
	fp32	fp32	Optional	
	fp16	fp16	Optional	
	sym-i16	sym-i16	Optional	
	bf16	bf16	Optional	

3 Fuse Operation Support

This chapter lists the operation combinations that NPU V9.2.0 supports for fusion. The NN engine of the NPU can fuse operations on the fly to reduce computation and bandwidth overhead.

In this chapter, the operation prefix VSI_NN_OP_ is omitted. Other abbreviations are as follows:

- DW_2D (abbr.) VSI_NN_OP_CONV2D where the multiplier is 1
- MAX_POOL (abbr.) VSI_NN_OP_POOL where the pool type is MAX
- HSWISH (abbr.) VSI_NN_OP_SWISH where the type is HSWISH

3.1 Fuse Operations

Table 9 lists the operation combinations that the NPU can fuse.

Note 1: The support for fuse operations may vary with software releases.
Note 2: Some operation combinations may not be supported by certain NPU hardware versions.
Note 3: In the table, operations in blue are either added or updated in this release.

Table 9. Fuse Operation Support with NPU

Fuse Operation	**First Operation**					
Second Operation	CONV2D	CONV1D	DW_2D (abbr.)	FCL2	MATRIXMUL	PERMUTE
ABS	Supported	Supported	Supported	Supported	Supported	
ACOSH	Supported	Supported	Supported	Supported	Supported	
ADD	Supported	Supported	Supported	Supported		
ATAN	Supported	Supported	Supported	Supported	Supported	
ATANH	Supported	Supported	Supported	Supported	Supported	
CELU	Supported	Supported	Supported	Supported	Supported	
CLIP	Supported	Supported	Supported	Supported	Supported	
CONV1D						Supported
CONV2D						Supported
COS	Supported	Supported	Supported	Supported	Supported	
DEPTH2SPACE	Supported	Supported	Supported	Supported		
DW_2D (abbr.)						Supported
ELU	Supported	Supported	Supported	Supported	Supported	
ERF	Supported	Supported	Supported	Supported	Supported	
EXP	Supported	Supported	Supported	Supported	Supported	
GELU	Supported	Supported	Supported	Supported	Supported	
HARD_SIGMOID	Supported	Supported	Supported	Supported	Supported	
HSWISH (abbr.)	Supported	Supported	Supported	Supported	Supported	
INVERSE_SIGMOID	Supported	Supported	Supported	Supported	Supported	
LEAKY_RELU	Supported	Supported	Supported	Supported	Supported	
LOG	Supported	Supported	Supported	Supported	Supported	
MAX_POOL (abbr.)	Supported	Supported	Supported	Supported		

Fuse Operation	**First Operation**					
Second Operation	CONV2D	CONV1D	DW_2D (abbr.)	FCL2	MATRIXMUL	PERMUTE
MAXIMUM	Supported	Supported	Supported	Supported		
MINIMUM	Supported	Supported	Supported	Supported		
MISH	Supported	Supported	Supported	Supported	Supported	
MULTIPLY	Supported	Supported	Supported	Supported		
NEG	Supported	Supported	Supported	Supported	Supported	
PERMUTE	Supported	Supported	Supported	Supported		
PRELU	Supported	Supported	Supported	Supported		
RCP	Supported	Supported	Supported	Supported	Supported	
RELATIONAL_OPS	Supported	Supported	Supported	Supported		
RELU	Supported	Supported	Supported	Supported	Supported	
RELUN	Supported	Supported	Supported	Supported	Supported	
RESHAPE	Supported	Supported	Supported	Supported	Supported	
ROUND	Supported	Supported	Supported	Supported	Supported	
RSQRT	Supported	Supported	Supported	Supported	Supported	
SELU	Supported	Supported	Supported	Supported	Supported	
SIGMOID	Supported	Supported	Supported	Supported	Supported	
SIGN	Supported	Supported	Supported	Supported	Supported	
SOFTRELU	Supported	Supported	Supported	Supported	Supported	
SOFTSIGN	Supported	Supported	Supported	Supported	Supported	
SPACE2DEPTH	Supported	Supported	Supported	Supported		
SQRT	Supported	Supported	Supported	Supported	Supported	
SQUARE	Supported	Supported	Supported	Supported	Supported	
SUBTRACT	Supported	Supported	Supported	Supported		
SWISH	Supported	Supported	Supported	Supported	Supported	
TAN	Supported	Supported	Supported	Supported	Supported	
TANH	Supported	Supported	Supported	Supported	Supported	
MAX_POOL (abbr.) + ABS	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + ACOSH	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + ADD	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + ATAN	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + ATANH	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + BATCH_NORM	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + CELU	Supported	Supported	Supported	Supported		

Fuse Operation	First Operation					
Second Operation	CONV2D	CONV1D	DW_2D (abbr.)	FCL2	MATRIXMUL	PERMUTE
MAX_POOL (abbr.) + CLIP	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + COS	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + ELU	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + ERF	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + EXP	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + GELU	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + HARD_SIGMOID	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + HSWISH	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + INVERSE_SIGMOID	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + LEAKY_RELU	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + LOG	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + MAXIMUM	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + MINIMUM	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + MISH	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + MULTIPLY	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + NEG	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + PRELU	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + RCP	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + RELATIONAL_OPS	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + RELU	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + RELUN (abbr.)	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + ROUND	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + RSQRT	Supported	Supported	Supported	Supported		

Fuse Operation	First Operation					
Second Operation	CONV2D	CONV1D	DW_2D (abbr.)	FCL2	MATRIXMUL	PERMUTE
MAX_POOL (abbr.) + SELU	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + SIGMOID	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + SIGN	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + SOFTRELU	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + SOFTSIGN	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + SQRT	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + SQUARE	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + SUBTRACT	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + SWISH	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + TAN	Supported	Supported	Supported	Supported		
MAX_POOL (abbr.) + TANH	Supported	Supported	Supported	Supported		

Document Revision History

This chapter describes the differences between different reversions of this document.

Note: In this table, the operation prefix `VSI_NN_OP_` is omitted.

Document Revision	Date	Compatible Software	Change History
0.52	2023-12-18	• Unified driver 6.4.17 and later • OVXLIB 1.2.2 and later	• Section 2.3, *Activation Operations* Added TAN. • Section 2.9, *Miscellaneous Operations* Added `CROP_AND_RESIZE`. • Section 3.1, *Fuse Operations* Added TAN and `MAX_POOL` + TAN as the second operation of fuse operations.
0.51	2023-09-19	• Unified driver 6.4.16 and later • OVXLIB 1.1.88 and later	• Section 2.6, *Reshape Operations*: Added `REDUCEL2`. • Section 3.1, *Fuse Operations*: Added support for `MATRIXMUL` as the first operation.
0.50	2023-06-28	• Unified driver 6.4.15 and later • OVXLIB 1.1.84 and later	Initial release for NPU hardware V9.2.0.